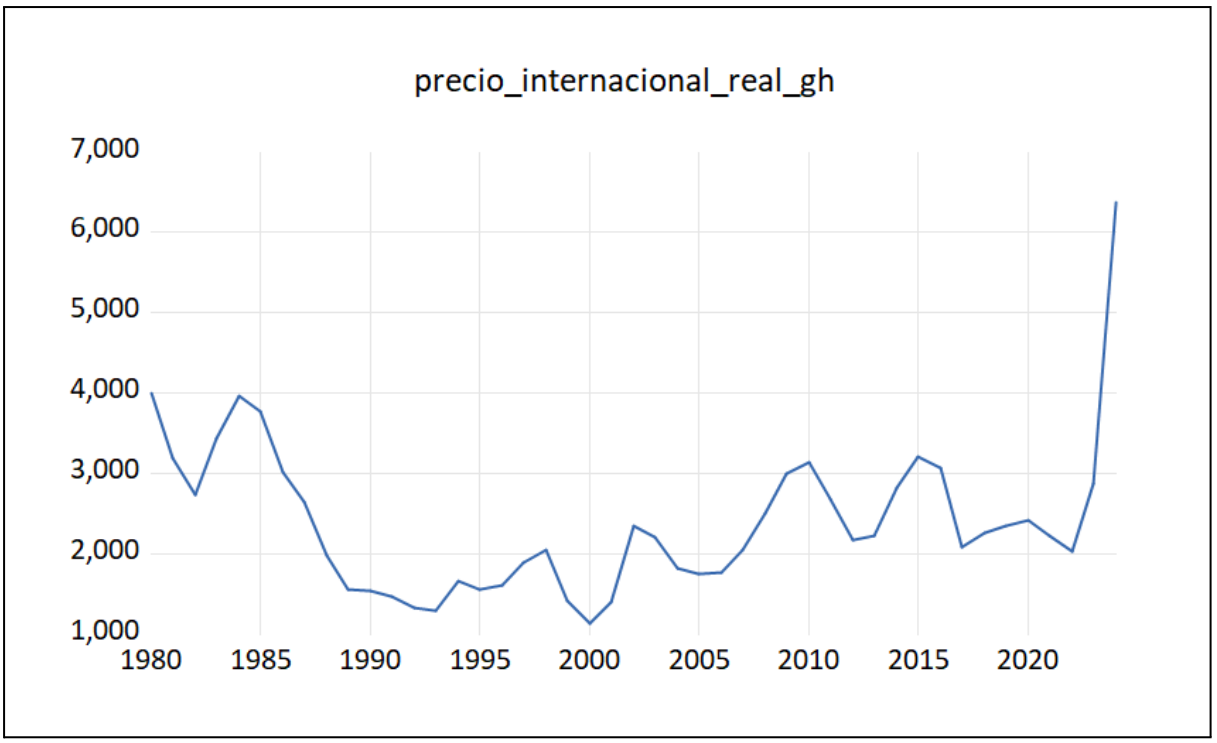


En el gráfico se observa una fuerte caída entre 1985 y 1990, luego cierta estabilidad, y finalmente una gran subida en los últimos años del precio internacional del cacao.

Analizo los test de Dickey Fuller para verificar la existencia de raíz unitaria, empezando con el modelo con tendencia determinística e intercepto:

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-1.243925	0.8882
Test critical values:	1% level	-4.186481	
	5% level	-3.518090	
	10% level	-3.189732	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:49
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	-0.172549	0.138713	-1.243925	0.2210
D(PRECIO INTERNACIONAL REAL GH	0.616396	0.226211	2.724873	0.0096
C	221.7287	385.0076	0.575907	0.5680
@TREND("1980")	11.30600	7.774911	1.454164	0.1539

R-squared	0.240057	Mean dependent var	74.03259
Adjusted R-squared	0.181600	S.D. dependent var	682.1569
S.E. of regression	617.1163	Akaike info criterion	15.77640
Sum squared resid	14852469	Schwarz criterion	15.94023
Log likelihood	-335.1926	Hannan-Quinn criter.	15.83682
F-statistic	4.106556	Durbin-Watson stat	1.119953
Prob(F-statistic)	0.012631		

Como la tendencia determinística no resulta significativa (p-valor = 0.1539), vuelvo a hacer el test con intercepto.

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: Constant
Lag Length: 2 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-0.566130	0.8673
Test critical values:	1% level	-3.596616	
	5% level	-2.933158	
	10% level	-2.604867	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:49
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	-0.081703	0.144319	-0.566130	0.5746
D(PRECIO INTERNACIONAL REAL GH	0.834834	0.227768	3.665281	0.0008
D(PRECIO INTERNACIONAL REAL GH	-0.628098	0.235939	-2.662122	0.0113
C	247.0291	340.6667	0.725134	0.4728

R-squared	0.316966	Mean dependent var	86.39314
Adjusted R-squared	0.263043	S.D. dependent var	685.5347
S.E. of regression	588.5057	Akaike info criterion	15.68344
Sum squared resid	13160881	Schwarz criterion	15.84894
Log likelihood	-325.3523	Hannan-Quinn criter.	15.74410
F-statistic	5.878048	Durbin-Watson stat	1.412468
Prob(F-statistic)	0.002123		

Nuevamente, el intercepto no resulta significativo (p-valor = 0.4728), por lo que realizo el test sin tendencia ni intercepto.

Null Hypothesis: PRECIO INTERNACIONAL REAL GH has a unit root
Exogenous: None
Lag Length: 2 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		0.496799	0.8186
Test critical values:	1% level	-2.621185	
	5% level	-1.948886	
	10% level	-1.611932	

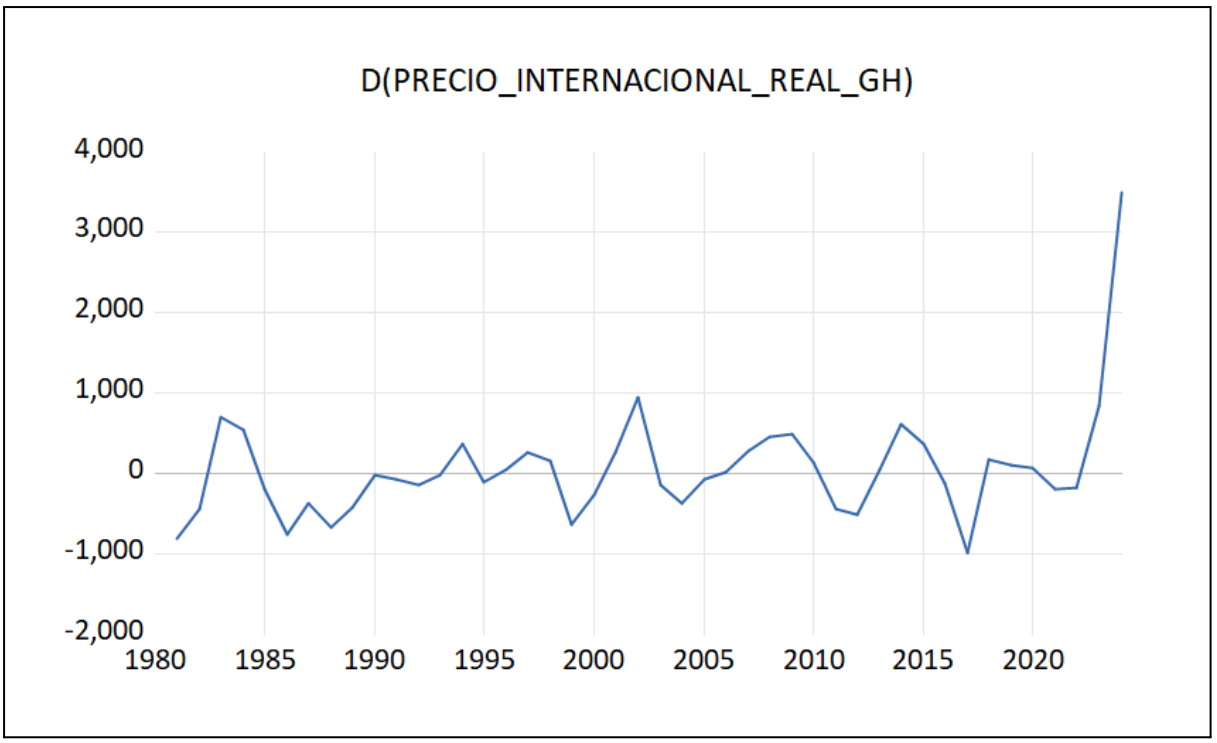
*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO INTERNACIONAL REAL GH)
Method: Least Squares
Date: 10/14/26 Time: 18:50
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO INTERNACIONAL REAL GH(-1)	0.019113	0.038472	0.496799	0.6221
D(PRECIO INTERNACIONAL REAL GH	0.802145	0.221901	3.614878	0.0008
D(PRECIO INTERNACIONAL REAL GH	-0.682925	0.222133	-3.074394	0.0038

R-squared	0.307515	Mean dependent var	86.39314
Adjusted R-squared	0.272003	S.D. dependent var	685.5347
S.E. of regression	584.9171	Akaike info criterion	15.64957
Sum squared resid	13342993	Schwarz criterion	15.77369
Log likelihood	-325.6409	Hannan-Quinn criter.	15.69506
Durbin-Watson stat	1.441096		

Como el p-valor asociado al test estadístico es 0.8186 no rechazo la hipótesis nula de raíz unitaria por lo que debo diferenciar la serie:



Notar la presencia de un valor atípico en los últimos años (2023-2024) donde el precio internacional se dispara.

Realizo el test de Dickey Fuller nuevamente para asegurarme de que no haya que diferenciar nuevamente la serie:

Null Hypothesis: D(PRECIO_INTERNACIONAL_REAL_GH) has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 1 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.690253	0.0341
Test critical values:	1% level	-4.192337	
	5% level	-3.520787	
	10% level	-3.191277	

*MacKinnon (1996) one-sided p-values.

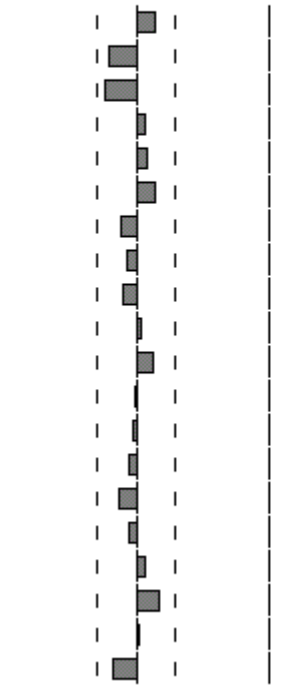
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_INTERNACIONAL_REAL_GH,2)
Method: Least Squares
Date: 10/20/26 Time: 17:58
Sample (adjusted): 1983 2024
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(PRECIO_INTERNACIONAL_REAL_GH(-1	-0.959795	0.260089	-3.690253	0.0007
D(PRECIO_INTERNACIONAL_REAL_GH(-1	0.716317	0.215317	3.326810	0.0020
C	-293.3396	192.6193	-1.522898	0.1361
@TREND("1980")	14.97777	7.259097	2.063311	0.0460

R-squared	0.314819	Mean dependent var	93.76005
Adjusted R-squared	0.260726	S.D. dependent var	651.7977
S.E. of regression	560.4226	Akaike info criterion	15.58565
Sum squared resid	11934794	Schwarz criterion	15.75114
Log likelihood	-323.2987	Hannan-Quinn criter.	15.64631
F-statistic	5.819927	Durbin-Watson stat	1.563827
Prob(F-statistic)	0.002247		

Como la tendencia resulta significativa y el estadístico del test tiene un p-valor = 0.0341, rechazo la hipótesis nula de raíz unitaria en la serie diferenciada, al 5% de significatividad.

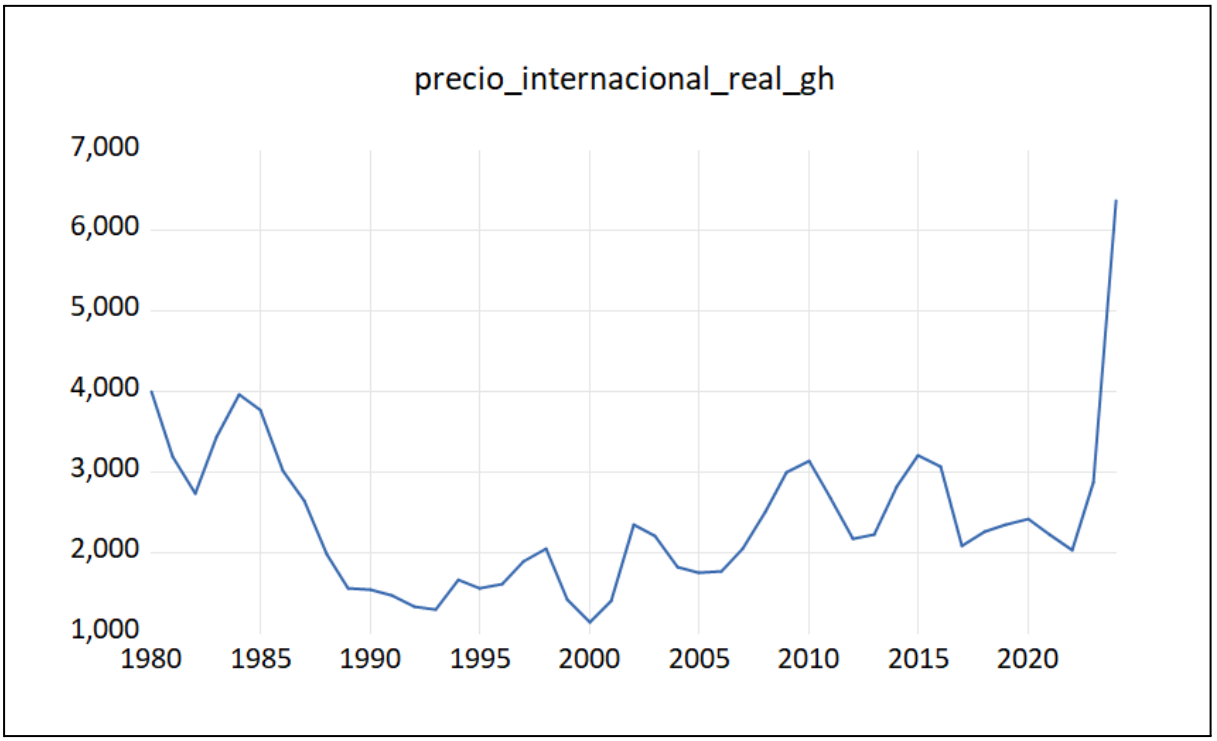
Date: 10/14/26 Time: 18:52
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.145	0.145	0.9644	0.326
		2	-0.219	-0.245	3.2214	0.200
		3	-0.245	-0.185	6.1186	0.106
		4	0.072	0.096	6.3733	0.173
		5	0.084	-0.035	6.7320	0.241
		6	0.148	0.144	7.8796	0.247
		7	-0.110	-0.122	8.5321	0.288
		8	-0.079	0.013	8.8812	0.352
		9	-0.099	-0.091	9.4395	0.398
		10	0.037	-0.017	9.5193	0.484
		11	0.133	0.120	10.582	0.479
		12	-0.000	-0.103	10.582	0.565
		13	-0.017	0.120	10.601	0.644
		14	-0.056	-0.072	10.807	0.701
		15	-0.137	-0.147	12.110	0.671
		16	-0.054	-0.016	12.319	0.722
		17	0.082	-0.035	12.816	0.748
		18	0.181	0.197	15.354	0.638
		19	0.019	-0.059	15.385	0.698
		20	-0.186	-0.075	18.284	0.569

El correlograma de la serie diferenciada se comporta como ruido blanco, lo cual tiene sentido teniendo en cuenta que se trata de la serie de precio internacional del cacao, que al ser una commodity es normal que se comporte como un *random walk*.

Incorporando una variable de intervención:

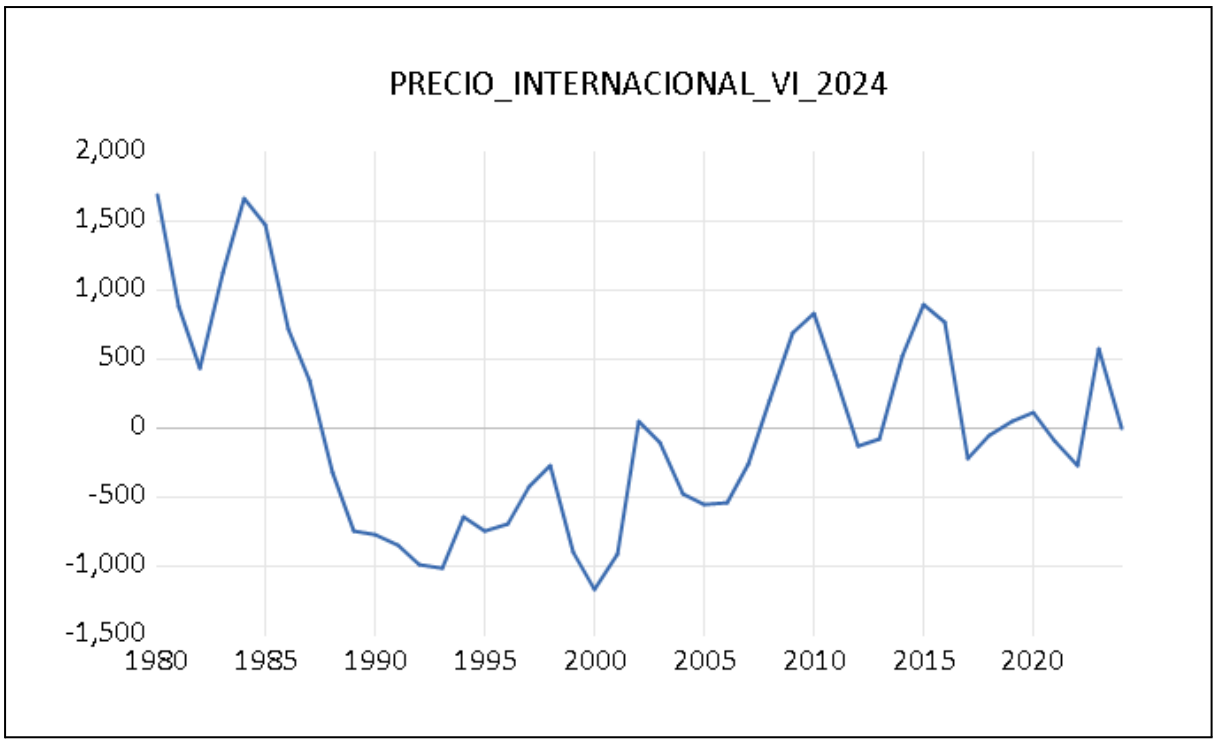


A pesar de que el modelo anterior reportaba un random walk, lo que tiene sentido económico teniendo en cuenta de que se trata de una commoditie, incorporo una variable de intervención debido al pico en el precio internacional en el año 2024.

Dependent Variable: PRECIO_INTERNACIONAL_REAL_GH
Method: Least Squares
Date: 11/02/26 Time: 12:49
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2309.389	111.9929	20.62085	0.0000
VI_2024	4058.679	751.2712	5.402415	0.0000

R-squared	0.404317	Mean dependent var	2399.582
Adjusted R-squared	0.390464	S.D. dependent var	951.5185
S.E. of regression	742.8769	Akaike info criterion	16.10236
Sum squared resid	23730240	Schwarz criterion	16.18266
Log likelihood	-360.3032	Hannan-Quinn criter.	16.13230
F-statistic	29.18608	Durbin-Watson stat	0.360362
Prob(F-statistic)	0.000003		



Null Hypothesis: PRECIO_INTERNACIOAL_VI_2024 has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-2.857434	0.0053
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

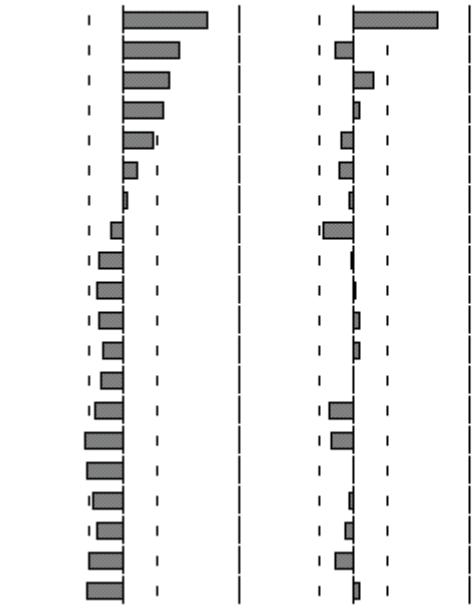
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO_INTERNACIOAL_VI_2024)
Method: Least Squares
Date: 11/02/26 Time: 13:00
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO_INTERNACIOAL_VI_2024(-1)	-0.239806	0.083923	-2.857434	0.0066

R-squared	0.153212	Mean dependent var	-38.23194
Adjusted R-squared	0.153212	S.D. dependent var	444.2701
S.E. of regression	408.8221	Akaike info criterion	14.88690
Sum squared resid	7186828.	Schwarz criterion	14.92745
Log likelihood	-326.5119	Hannan-Quinn criter.	14.90194
Durbin-Watson stat	1.412971		

Ahora el test de Dickey Fuller rechaza la hipótesis nula de raíz unitaria, al contrario que el modelo anterior sin variable de intervención.

Date: 11/02/26 Time: 13:01
Sample: 1980 2024
Included observations: 45



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.760	0.760	27.778	0.000
		2	0.508	-0.165	40.483	0.000
		3	0.402	0.188	48.620	0.000
		4	0.362	0.047	55.361	0.000
		5	0.270	-0.104	59.204	0.000
		6	0.129	-0.119	60.106	0.000
		7	0.028	-0.030	60.149	0.000
		8	-0.113	-0.270	60.872	0.000
		9	-0.215	-0.018	63.592	0.000
		10	-0.240	0.014	67.076	0.000
		11	-0.211	0.053	69.833	0.000
		12	-0.183	0.049	71.970	0.000
		13	-0.189	-0.004	74.343	0.000
		14	-0.254	-0.216	78.742	0.000
		15	-0.344	-0.197	87.068	0.000
		16	-0.336	-0.000	95.286	0.000
		17	-0.265	-0.026	100.59	0.000
		18	-0.240	-0.069	105.09	0.000
		19	-0.314	-0.161	113.12	0.000
		20	-0.332	0.056	122.42	0.000

Dependent Variable: PRECIO_INTERNACIOAL_VI_2024
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 11/02/26 Time: 13:02
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments
Convergence achieved after 11 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.760194	0.083923	9.058187	0.0000

R-squared	0.655078	Mean dependent var	-38.23194
Adjusted R-squared	0.655078	S.D. dependent var	696.1040
S.E. of regression	408.8221	Akaike info criterion	14.88690
Sum squared resid	7186828.	Schwarz criterion	14.92745
Log likelihood	-326.5119	Hannan-Quinn criter.	14.90194
Durbin-Watson stat	1.412971		

Inverted AR Roots	.76

Date: 11/02/26 Time: 13:02
Sample (adjusted): 1981 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.267	0.267	3.3551	
		2	-0.224	-0.318	5.7717	0.016
		3	-0.290	-0.151	9.9254	0.007
		4	-0.006	0.078	9.9269	0.019
		5	0.110	-0.023	10.561	0.032
		6	0.006	-0.071	10.563	0.061
		7	0.136	0.245	11.571	0.072
		8	0.015	-0.114	11.584	0.115
		9	-0.126	-0.088	12.496	0.130
		10	-0.143	0.011	13.711	0.133
		11	0.016	-0.002	13.727	0.186
		12	0.127	-0.000	14.744	0.194
		13	0.167	0.198	16.569	0.167
		14	-0.019	-0.132	16.593	0.219
		15	-0.255	-0.205	21.141	0.098
		16	-0.201	0.028	24.056	0.064
		17	0.031	-0.006	24.130	0.087
		18	0.182	-0.027	26.715	0.062
		19	-0.047	-0.119	26.891	0.081
		20	-0.185	-0.127	29.789	0.055

Dependent Variable: PRECIO_INTERNACIOAL_VI_2024
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 11/02/26 Time: 13:03
Sample (adjusted): 1982 2024
Included observations: 43 after adjustments
Convergence achieved after 11 iterations
Coefficient covariance computed using outer product of gradients

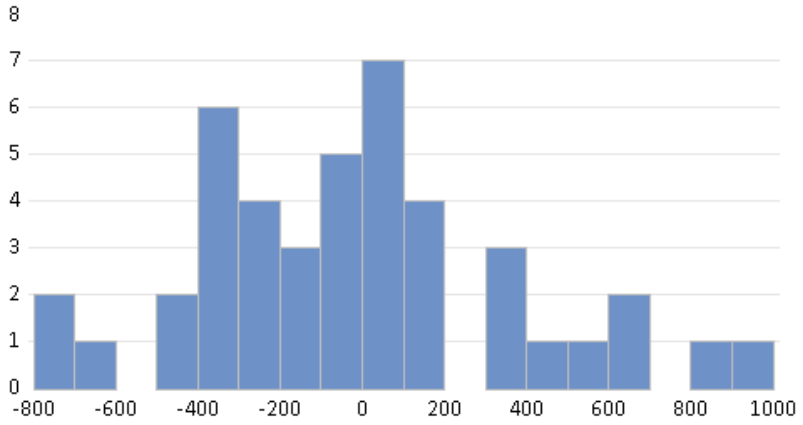
Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	1.039179	0.148900	6.979024	0.0000
AR(2)	-0.285599	0.140693	-2.029946	0.0489

R-squared	0.681592	Mean dependent var	-59.47634
Adjusted R-squared	0.673826	S.D. dependent var	689.7585
S.E. of regression	393.9323	Akaike info criterion	14.83563
Sum squared resid	6362490.	Schwarz criterion	14.91755
Log likelihood	-316.9661	Hannan-Quinn criter.	14.86584
Durbin-Watson stat	1.732396		

Inverted AR Roots	.52-.13i	.52+.13i

Date: 11/02/26 Time: 13:03
Sample (adjusted): 1982 2024
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.098	0.098	0.4438	
		2	-0.145	-0.156	1.4328	
		3	-0.164	-0.137	2.7383	0.098
		4	0.089	0.103	3.1351	0.209
		5	0.115	0.057	3.8096	0.283
		6	-0.092	-0.115	4.2501	0.373
		7	0.109	0.195	4.8883	0.430
		8	0.009	-0.034	4.8929	0.558
		9	-0.107	-0.137	5.5411	0.594
		10	-0.153	-0.071	6.9223	0.545
		11	-0.018	-0.028	6.9430	0.643
		12	0.041	-0.072	7.0500	0.721
		13	0.139	0.182	8.2948	0.687
		14	-0.023	-0.047	8.3316	0.759
		15	-0.239	-0.248	12.266	0.506
		16	-0.178	-0.077	14.542	0.410
		17	-0.007	-0.031	14.546	0.485
		18	0.140	-0.014	16.053	0.449
		19	-0.088	-0.093	16.671	0.477
		20	-0.176	-0.165	19.269	0.375



Series: Residuals
Sample 1982 2024
Observations 43

Mean	-22.58094
Median	-25.69318
Maximum	930.7882
Minimum	-763.0330
Std. Dev.	388.5432
Skewness	0.427440
Kurtosis	3.012949
Jarque-Bera	1.309684
Probability	0.519524